

DentalVerse: Interactive Multiusers Virtual Reality Implementation to train preclinical dental student psychomotor skill

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ABSTRACT

The COVID-19 pandemic created the largest disruption of education systems in history. Distance learning through online platforms were part of the solution. However, preclinical exercises to train psychomotor skills of learners have been challenging. The use of virtual reality (VR) in training medical students is innovative and has attracted much attention. In this study, authors presented the development of multi-user VR application for dental education. Our preliminary results showed the potential of using VR in the preclinical curriculum of dental education.

Keywords: Immersive learning, multiusers, virtual reality, multiusers.

Index Terms: K.6.1 [Management of Computing and Information Systems]: Project and People Management—Life Cycle; K.7.m [The Computing Profession]: Miscellaneous—Ethics

1 INTRODUCTION

The COVID-19 pandemic has caused major lockdown in many countries. In the United States, several states implemented a stay-at-home policy, causing many educational institutions, particularly dental schools, to close and suspend their preclinical curricular and clinical activities. The sudden halt of academic activities in these dental schools has made significant changes in the pedagogy approach. American Dental Education Association has reported substantial educational reforms, such as shifting in-person didactic courses to online platforms where classes might be delivered in asynchronous or synchronous forms. [1].

The usage of online learning has vastly increased in higher education especially since the COVID-19 pandemic outbreak as it has no time and space limitations. In this era of information technology, the internet provides ubiquitous digital information regarding education materials and teaching videos that will ease students' understanding specifically in dental subjects. However, the success of e-learning depends on many factors, which are the attitudes and interactive teaching styles of the faculty, as well as on the experience and attitudes of students with regard to technology [16]. As a result, many educators, including university faculty, opted to transition their courses to live synchronous web meetings using web conferencing tools like Zoom and Webex. One of the advantages of live communication tools over in-person classes is that they enable lecturers to provide presentations even if the participants are located in different cities, states, or even countries [5].

However, this learning platform may have some limitations such as restricted interaction and 2D space.

Traditionally, dental curriculum relies heavily on psychomotor trainings, starting in preclinical activities before matriculating to the clinical settings. For example, a dental student hones their hand skill through typodonts and manikins simulation laboratory exercise during the first 2 years in the preclinical curriculum. Given the current pandemic crisis, dental institutions explore alternative options to prepare students with these psychomotor skills in a safe and viable environment. Without a doubt, teaching preclinical simulation exercise during this pandemic has been challenging. An additional measure, such as social distancing, has been proposed by the public health officials to slow the virus's spread. This mandate may prohibit the instructors-students from interacting closely and providing instantaneous assessment and feedback.

2 PREVIOUS WORKS

The use of virtual simulation in training health care professionals has attracted much attention and offered opportunities in teaching and learning. Operating room performance and self-confidence levels have improved after practicing with a virtual reality simulator [10][14]. Also, a 3D virtual learning space cultivates student learning and results in a similar understanding when compared to a traditional face-to-face classroom setting [6]. Many dental virtual simulation studies were conducted using the Simodont (developed by the Academic Centre for Dentistry Amsterdam (ACTA), Amsterdam, Netherlands). Simodont is a sensitive tool that can detect performance differences between novice and more experienced students [7]. Simodont could also improve the student's satisfactory performance [8].

Virtual Reality (VR) has been more commonly applied in the oral surgery field. For example, using VR as a training tool for an orthognathic surgery [11]. Participating expert oral surgeons indicated VR surgery could be used as a valid training instrument [11]. In a randomized control trial, the efficacy of VR was evaluated, and its use provided an alternative treatment to manage dental anxiety and dental phobia [12]. Further, most of the described instruments used in the studies, were operated as a single-user supplement material. This gives a limited benefit for dental education, as feedback and mentorship are important part of student learning. Therefore, instead of an isolated single-user experience, it is essential to provide a communal experience inside the VR application by allowing multiple users to meet and interact in the same computer-generated environment. This study aimed to describe the development of an immersive, multi-user VR program for dental implant surgery training. The main motivation of this project is to demonstrate the use of immersive, multi-user VR to deliver teaching and learning activities with multiple modalities including PowerPoint and whiteboard functionality. This project also showed the possibility of a multi-user VR application for academic exam purposes.

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With the impact COVID-19 has had on education, utilizing different methods of virtual learning has become a vital resource in replicating the same level of engagement experienced through in-person lectures. The College of Dentistry, at the University of Illinois in Chicago, is working towards the implementation of VR learning into standard lesson plans for first-year dental students.

Continuing development from the same application, our team expanded the program's features further by incorporating additional features available to the professor and students that promote student engagement. VR creates an entirely immersive digital 360 environment. These environments have also been shown to improve recall and retention rates in students [4]. Faculty and students are able to interact with each other in the virtual world, through the multi-user function hosted in two environment options. Supporting immersive learning, users have access to different features that support immersive learning, these include slide presentations, PDFs, interactable whiteboards, and a student assessment room.

3 DENTALVERSE PROJECT

3.1 The Environment

The virtual environment consisted two main rooms: classroom (figure 1) and dental clinic (figure 2). The overall design is adapted from the real physical space. The classroom has desks, chairs, personal whiteboards, and a screen for educational slides. Dental treatment room consists of the essential elements including an operator chair and dentistry tools. We also added several everyday objects one would expect to see when visiting a dental clinic such as furniture and cabinets.

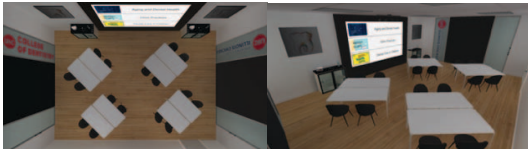


Figure 1: DentalVerse's class room

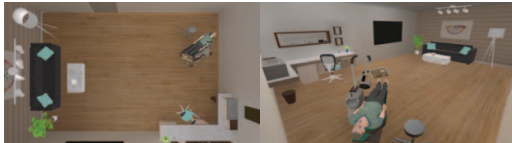


Figure 2: DentalVerse's dental clinic

There are several points to take into consideration when deciding if a VR educational space is beneficial. Because first-year dental students are the intended users of this application, VR offers several opportunities that could potentially not be available to them otherwise. VR gives them the ability to access experience that may otherwise be untenable due to patient liability from mistakes that could occur in the real world, these mistakes could also lead to the student's own fall in self-esteem [9].

Taking these points into consideration, the College of Dentistry at the University of Illinois wanted to further expand the environment's functionality to include a classroom setting for students to be able to engage in a lecture-style class while maintaining the benefits of VR learning. Due to Multiplayer being a leading feature in the application, it was important to layout a space that was easy to maneuver in VR, while also encouraging interaction among peers. The traditional grouped desking layout not only promotes student interactions but allows users to have ease of access to multiple vantage points when looking throughout the room in VR.

Additional features within the lecture room environment that were included were the vertical University of Illinois Chicago

College of Dentistry name and logo banners on the windows along the sides of the room. These were used to reinforce the sense of place while immersed in VR. On the walls of either side of the room, there are also placeholders for the professor and other users to display the Slide Presentation, PDF, and Whiteboard attributes offered in the application. Paired with the school's branding in the scene, modelled props were included to blend the two mediums. These assets included VR headsets and various dental instruments, featuring dentures, excavators, tweezers, and mouth mirrors

Lighting conditions in VR educational spaces can have a major impact on the student's cognitive functionality. High luminance light settings in a VR environment have been shown to enhance students' long memory retention [15]. However, maintaining high luminance can also cause discomfort to the users over long spans of time. To counter the effect this could pose on the user, the team plans to work on improving the lighting quality throughout and find acceptable mid-range lighting levels by conducting user tests to find adequate levels of brightness within each environment while also utilizing a combination of real time and baked lights to improve overall rendering quality without posing a drop in performance.

3.2 User Features

As a part of the consideration for VR to be a profitable educational tool, the content must hold equivalent value and motivation within the virtual environment [9]. Therefore, if this educational simulation is to imitate and enhance a real-world classroom, the virtual must contain similar elements to the real; these include slide presentations, research papers, bulleted notes, and assessments.

3.2.1 Wrist UI

The wrist user-interface (UI) provides access to two features: the whiteboard and PDF file user. By facing their palms upwards, a user gains the ability to view this UI, which provides controls for the whiteboard and PDF. As seen in figure 3, the main button (E) expands the interface's options, avoiding screen clutter if the user is simply moving their wrist. Upon expansion, the representative icons (C) and (D) display or hide their corresponding features, ensuring correct orientation facing the player based on the head-mounted display's direction.

Interaction with the wrist UI occurs through a finger tap such as a press of a phone's touchscreen. Because virtual reality currently lacks the feeling of pressing a physical screen, haptic feedback in the form of light vibration fills this void. Additionally, each icon which acts as a feature toggle changes coloration depending on its current status. The physical and visual indicators of selection have been implemented with the goal of providing a realistic experience to the user.

The wrist UI also provides a sense of individuality, matching that felt in a real-world classroom. While students bring notebooks and laptops to in-person classes, the application provides virtual students a personal means to write, draw diagrams, and study research documents.



Figure 3: Wrist user-interface. (A) Locking, (B) resetting, and (C) enabling the whiteboard. (D) Enabling the PDF display. (E) Expanding the menu.



Figure 4: The PDF display with a touch-enabled user-interface.

3.2.2 Whiteboard

Professors and students are allowed to use a virtual whiteboard (figure 5) for teaching, note-taking, and brainstorming. To provide a smooth and practical virtual writing experience, the whiteboard's markers mimic those of the real-world. The plane of the whiteboard is an image. A short, invisible line is drawn from a marker's tip to the closest pixel of this plane, which triggers the image's pixels to alter color in that area.

Using virtual reality to its advantage, the whiteboard component has great flexibility in its usage; the features that allow for this include the whiteboard enable/disable functions, lock/unlock functions, and reset functions. The buttons corresponding to these functions can be found on the wrist interactive interface. With the enable/disable function, the whiteboard will appear/disappear upon the touch of the whiteboard icon. In the case of the lock/unlock, the whiteboard is unlocked by default when first enabled. Until locked, the whiteboard follows the user's forward direction through the head-mounted display's camera. Once locked, the board is stationary in the last unlocked position. With the reset button, a user can erase the whiteboard instantly.

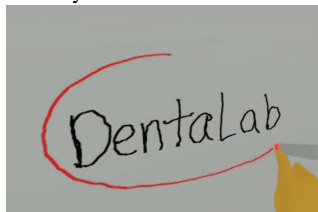


Figure 5: A user writing on the whiteboard with multiple colors.

3.2.3 Slide Presentations

In the Lecture Room environment, a place for the professor to present slide presentations sits in the front of the room. The controls are located at various points in the room; these include two points in the front of the room and two areas on opposing walls in the back.

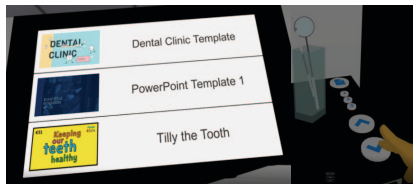


Figure 6: Slide presentation and controller button.

3.2.4 Student Assessment

Utilizing an existing exam room, from the previous project, the team transformed it to serve as a space for student users to take exams while maintaining immersion in the VR world. A student may access the assessment room by grabbing the doorknob; this interaction matches a real-world expectation. The room offers the user a place to take multiple-choice tests in VR while eliminating outside distractions that occur in the real-world environment. This feature was implemented using an outside .txt document while

also being able to log the student's answer choices. The exam is applied to a scrolling canvas in the scene, easing navigation should the student need to review a previous question. On the last question, a student is able to submit their work for automatic evaluation, where they can compare their answers with those correct and see their final score. The formatting of the text file provides the way for new exams to be administered. With no code modification necessary, the professor can follow this formatting to edit, add, or remove exam questions and answers within the text document itself.

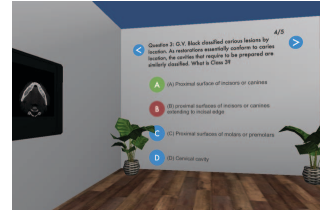


Figure 7: The exam assessment room looking through a submitted assignment.

3.2.5 Multiusers VR Experience

Versions of VR learning began to be implemented in educational settings through head-mounted displays from as far back as the 1960s to assist in flight simulations for the United States Air Force [4]. Similar ideas have been used in the medical school to allow for otherwise complex concepts and abstract thinking to be visualized by students in a setting that is safe to practice practical skills and visualize 3D anatomy for an understanding of scale and relation [2]. To enhance dentistry education, simulations of dental implant surgeries have seen successful development for use in producing student confidence and familiarity in the procedure [13]. Expanding on this work's multiplayer functionality, the team has built a collaborative experience for professors and students in an updated exam room and virtual lecture room.

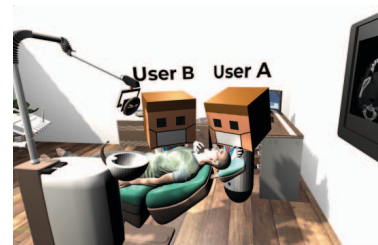


Figure 8: Multiusers VR dental training.

For multi-user implementation, the team integrated Photon Unity Networking 2. Remote procedure calls send local data from one client to all connected users; this ensures the synchronization of objects (e.g. surgical tools, patient mouth) and displays (e.g. slide presentation, whiteboard) in either multi-user room.

To provide seamless synchronization, a need arises for optimization. Low latency between clients contributes to the seemingly real-world interactions one can have in a VR environment; this latency requirement applies to the connection between the virtual world and its corresponding VR Display as well as the stability of each user's remote connection to one another. Prior to optimization, sending a high number of packets was necessary for all critical information to be shared. Unfortunately, receiving transmitted data of this magnitude led to delays in viewing the actions of other users through a considerable drop in frame rate. Low motion-to-photon (MTP) latency of 10 or less is recommended for VR applications to prevent users from suffering motion sickness. Higher MTP values can lead to motion sickness due to a disconnect between the human eye and a lapse

between the application's pixel's frame rate from movement [3]. Therefore, the development process required clever care into the prevalence of packets sent across the network.

Local data storage provides the greatest advantage in combating the number of packets brought into the application. For instance, since all data in PDF and PowerPoint type formats are static, small data types can be sent across the network to synchronize users. When a string and integer can represent the PowerPoint file path and its current slide, sending an entire byte array of image data is unnecessary. A remote procedure call notifies the other clients to locally run a specified function, which utilizes integer indexing and file paths to alter the local PowerPoint to match that of the remote.

Another minimization of packets occurs in the whiteboard implementation. In the background, the whiteboard is a Unity Texture2D object, which acts like a photograph whose pixels are editable. Rather than transmit an entire image upon change, each client can edit those pixels themselves through tracking whiteboard marker locations in real-time; this prevents the necessity of uploading and downloading the whiteboard's image. To provide a stable connection while in the application, only updating the room during real-time interactions is imperative.

4 CONCLUSION & FUTURE DIRECTION

COVID-19 showed an urgency of distance or online learning to support teaching and learning activities; however, conventional online learning platforms could not deliver 'hands-on' skill-related learning session. Immersive learning using VR is expected to tackle the challenges of delivering psychomotor skill-related course via conventional online platform. Furthermore, additional solutions are required to increase the efficiency of teaching and learning session inside the VR environment. In this paper, authors introduced **DentalVerse: Interactive Multiusers VR Implementation to train preclinical dental student's psychomotor skill**. In the DentalVerse project, we implemented a VR experience for cavity preparation training. Additionally, we integrated a multi-user feature that allows multiple users to join the same VR room to have a real time interaction. With this functionality, a professor or instructor can meet and interact with their students in real time from a distance. To support teaching and learning activities, DentalVerse is also equipped with other supporting modalities, including a virtual whiteboard and exam assessment room.

In the future, the authors plan to enhance DentalVerse project with other features, including a personalized avatar specific to each authorized user that important for security reason. Personalized avatars are also essential to establish trust between users inside VR environment. We also would like to exercise a possibility to use blockchain to track each student's performance and provide transparency even though all learning processes happened virtually. The authors also see potential integration with haptic devices, especially if this project is expanded into another dentistry direction that requires a haptic / tactile feedback as part of the knowledge transfer process. Lastly, the authors plan to conduct a user study to provide insight on student and faculty perceptions of the application.

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